NORMALISATION

It is used to organize the data into database.
It is used to reduce the data redundancy.
It is allowed to eliminate the insertion, updation, deletion anomalies (problem)

Types of Normalisation

1. 1NF - First Normal Form
2. 2NF - Second Normal Form
3. 3NF - Third Normal Form
4. BCNF - Boyce codd Normal Form

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1. 1NF (First Normal Form)

If a table want to be in 1NF first it should satisfy the following condition.

(i) The column name of the table should be unique and related

(ii) The domain of the table should be same

(iii) There should not be any multi valued attribute

2. 2NF (Second Normal Form)

If we want to perform 2NF first it should satisfy:

(i) All the conditions available in 1NF

(ii) There should be functional dependency among the attribute.

Note: fanctional dependancy is all the non prime attribute should be depend on prime attribute

3. 3NF (Third Normal Form)

If a table wanted to be in 3NF is should satisfy:

(i) All the conditions of 2NF

(ii) There should be transitive dependancy among the attributes

Note: An indirect dependancy among the

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column is called transitive dependency

4. BCNF (Boyce codd Normal Form)

It is also known as 3.5NF. It is an advanced version of 3NF. The anomalies/ problems resolve will be resolving in BCNF directly

Advantage of Normalization

- Reduce Duplicate Data
- Improves Data Integrity
- Saves energy
- Easier maintainance
- Avoids Anomalies

Disadvantage of Normalization

- More tables
- Complex Joins
- Query performance may reduce sometimes

Types of Normal Forms:

1. 1NF (First Normal Form) Rules:

- No multiple values in one column
- Each column contains atomic values
- No repeating groups.

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2. 2NF (Second Normal Form) Rules :

- Must be in 1NF
- No partial dependency
- Non-key attribute should depend on the whole primary key

3. 3NF (Third Normal Form) Rules :

- Must be in 2NF
- No Transitive dependency
- Non-key columns should depend only on the primary key

1NF :

Biscuit

Sid	Sname	Course
1	Raj	C, SQL
2	Rio	Java, C++
3	Ram	C#, SQL

---->

Student

Sid	Sname	C1	C2
1	Raj	C	SQL
2	Rio	Java	C++
3	Ram	C#	SQL

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2NF :

Student

sid	Sname	Pid	Pname
01	Raj	01	Ram
02	Rio	01	Ram
03	Raja	01	Ram

Student

sid	Sname	Pid
01	Raj	01
02	Rio	01
03	Raja	01

Professor

Pid	Pname
01	Ram

3NF :

Student

sid	Sname	Pid	Pname	Sage
01	Raj	01	Ram	18
02	Rio	01	Ram	17
03	Raja	01	Ram	20

Student

sid	Sname	Sage	Pid
01	Raj	18	01
02	Rio	17	01
03	Raja	20	01

Professor

Pid	Pname
01	Ram

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BCNF :

Student

Sid	Sname	Pname
01	Raj	Ram
02	Rio	Ram
03	Raja	Ram

Student

sid	Sname	Pid
01	Raj	01
02	Rio	01
03	Raja	02

Professor

Pid	Pname
01	Ram
02	Rami

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